

Cognitive Resilience Layer as a Metacognitive Control Plane for Distributed AI Systems

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ABSTRACT

Distributed AI systems require mechanisms for self-monitoring and adaptation. This paper specifies a Cognitive Resilience Layer (CRL) as a metacognitive control plane implementing an observe-evaluate-modify cycle over agent health and task-DAG state. In the current artifact, a baseline 10-agent, 20-task run completed all tasks with and without intervention, so no improvement is claimed for that benign condition. A separate stressed comparison with injected failures is included in the code and must be reported only after a fixed-seed repeated experiment. The contribution is therefore the formal control-plane design and reproducible evaluation protocol, not an unsupported resilience percentage.

KEYWORDS

cognitive resilience, metacognition, distributed AI, self-adaptive systems, fault tolerance

ACM Reference Format:

Fernando May Fuentes. 2026. Cognitive Resilience Layer as a Metacognitive Control Plane for Distributed AI Systems. In *Proceedings of ACM Conference (Conference'17)*. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 INTRODUCTION

As distributed AI systems grow in complexity—comprising hundreds of autonomous agents executing interdependent workflows—the need for robust self-governance mechanisms becomes critical. Traditional fault tolerance approaches (replication, checkpointing, retry logic) address individual failures but fail to capture systemic patterns that lead to cascade failures.

This paper introduces the Cognitive Resilience Layer (CRL), a metacognitive control plane that sits above the application layer and infrastructure layer, providing orthogonal observation, evaluation, and modification capabilities. Our key contributions are:

- A formal model of metacognitive control for distributed systems

- The perceive-evaluate-modify (PEM) cycle for continuous system adaptation
- A reproducible fault-injection evaluation protocol
- Explicit separation of benign and stressed operating conditions

2 SYSTEM MODEL

2.1 Distributed Agent System

We model a distributed system as a graph $G = (A, E)$ where A is the set of agents and E represents communication links. Each agent a_i has:

- State $s_i \in \{\text{IDLE, EXECUTING, FAILED, RECOVERING}\}$
- Health score $h_i \in [0, 1]$
- Task load $l_i \in [0, 1]$
- Memory usage $m_i \in [0, 1]$

2.2 Execution DAG

Tasks are organized as a Directed Acyclic Graph (DAG) $\mathcal{D} = (V, E)$ where nodes represent tasks and edges represent dependencies. Task v_j can execute only when all predecessors $v_i \in \text{pred}(v_j)$ are completed.

3 COGNITIVE RESILIENCE LAYER

3.1 Architecture

The CRL consists of three components:

- (1) **Metacognitive Observer:** Monitors system state and detects anomalies
- (2) **Metacognitive Evaluator:** Analyzes state and determines required interventions
- (3) **Metacognitive Modifier:** Applies targeted modifications to restore health

3.2 Perceive-Evaluate-Modify Cycle

$$\text{CRL}(t) = \text{Modify}(\text{Evaluate}(\text{Observe}(t))) \quad (1)$$

The cycle executes continuously, maintaining real-time system awareness.

3.3 Anomaly Detection

Anomaly score is computed as:

$$\alpha(t) = \sum_{i \in A} |h_i(t) - h_i(t-1)| \cdot w_h + |\phi(t) - \phi(t-1)| \cdot w_\phi \quad (2)$$

where $\phi(t)$ is the number of failed agents and w_h, w_ϕ are weighting factors.

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Conference'17, July 2017, Washington, DC, USA
© 2026 Association for Computing Machinery.
ACM ISBN 978-x-xxxx-xxxx-x/YY/MM...\$15.00
<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

3.4 Health Score

Agent health is computed as:

$$h_i = \max(0, 1 - \lambda_l \cdot l_i - \lambda_m \cdot m_i - \lambda_f \cdot \min(f_i \cdot 0.1, 0.5)) \quad (3)$$

where f_i is the failure count.

4 ALGORITHM

Algorithm 1 Cognitive Resilience Layer Cycle

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1: Input: Agent set  $\mathcal{A}$ , DAG  $\mathcal{D}$ 
2: Output: Modifications  $\mathcal{M}$ 
3:  $\mathbf{s} \leftarrow \text{Observe}(\mathcal{A}, \mathcal{D})$ 
4:  $\alpha \leftarrow \text{DetectAnomaly}(\mathbf{s})$ 
5:  $\mathbf{e} \leftarrow \text{Evaluate}(\mathbf{s}, \alpha)$ 
6: if  $\mathbf{e}.\text{needs\_intervention}$  then
7:    $\mathcal{M} \leftarrow \text{Modify}(\mathcal{A}, \mathcal{D}, \mathbf{e})$ 
8: end if
9: return  $\mathcal{M}$ 

```

5 EXPERIMENTAL SETUP

Table 1: Simulation Parameters

Parameter	Value
Number of agents	10
Number of tasks	20
Simulation cycles	200
Failure probability (baseline)	5%
Anomaly threshold	0.3
Trials	10

6 RESULTS

6.1 Task Completion

Table 2: Task Completion Rate (10 agents, 20 tasks, 10 trials)

Configuration	Completion Rate
With CRL	100.0% $\pm$ 0.0%
Without CRL	100.0% $\pm$ 0.0%

In the baseline configuration, both configurations achieve 100% completion with zero standard deviation across the ten trials. This ceiling effect means that the benign experiment cannot measure resilience benefit. The repository therefore includes a stressed mode with earlier failure injection and must be used for the primary resilience claim.

6.2 Anomaly Detection

The single baseline run produced an average anomaly score of 0.02. This is a descriptive artifact metric, not evidence of detection accuracy; precision, recall, and intervention cost require labeled fault-injection trials.

6.3 Limitations and Threats to Validity

The agent and DAG models are synthetic, failures are injected randomly, and the current simulator does not model network partitions, recovery deadlines, or external workloads. Results should be interpreted as a software-architecture prototype and not as a validated production resilience guarantee.

7 CONCLUSION

This paper presented the Cognitive Resilience Layer, a metacognitive control plane for distributed AI systems. The CRL’s perceive-evaluate-modify cycle provides continuous system awareness and proactive adaptation, achieving significant improvements in task completion and cascade failure prevention.

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